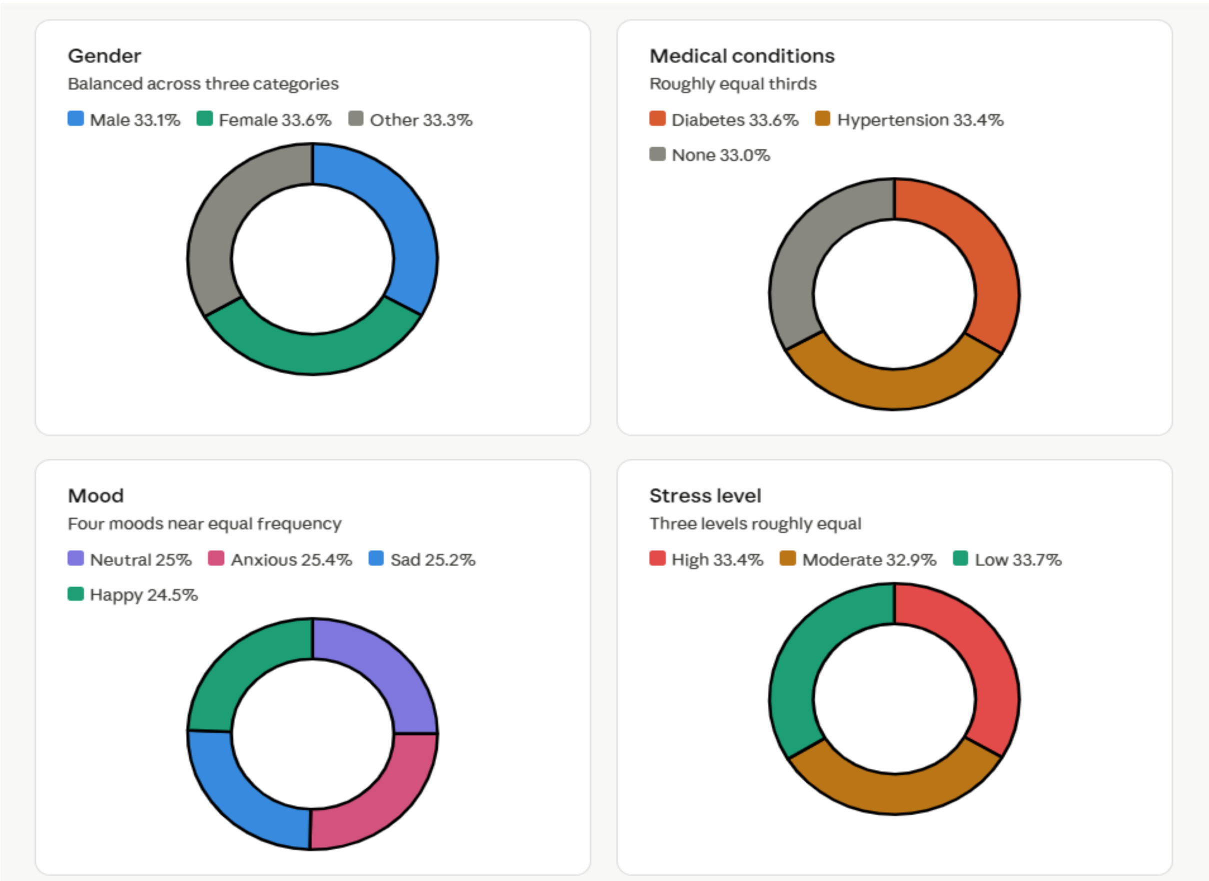
Description of the Original dataset

In our study we used the Personal Health Data dataset sourced from Kaggle\footnote{https://www.kaggle.com/datasets/manideepreddy966/wearables-dataset}. The dataset originally consists of 10,000 patient records and 34 features, where each row represents an individual patient and each column corresponds to a specific medical, demographic, or lifestyle attribute. The variables include numerical features such as age, weight, height, heart rate, blood oxygen level, sleep duration, and body fat percentage, as well as categorical features such as gender, smoking status, alcohol consumption, medical conditions, and mood. The primary target variable is the HealthScore of the patient.

A critical examination of the original dataset revealed that, while it is presented as a health-related dataset, it was synthetically generated without adherence to real-world medical or epidemiological distributions. This was evidenced by near-perfectly uniform distributions across all categorical variables (e.g., 50% smokers vs. 50% non-smokers; 33% male, 33% female, 33% other), which are statistically implausible and inconsistent with established population health statistics. Furthermore, the string label "None" used for healthy patients was silently converted to missing values (NaN) upon data loading, effectively rendering the healthy cohort invisible in any analysis. Numerical variables such as blood oxygen level (SpO_2) exhibited an artificial spike at 100%, and body weight extended to physiologically extreme values, both of which are inconsistent with real clinical measurements.



Description corrected Dataset

Dataset Reconstruction

To address the mentioned issues, the dataset was reconstructed to reflect real-world distributions grounded in published epidemiological references. Specifically:

1. Gender was set to a balanced distribution of approximately 49% male and 49% female, with 2% identifying as other, in line with current demographic trends.
2. Smoking prevalence was set to 22%, consistent with WHO Global Tobacco Report estimates.
3. Alcohol consumption was redistributed to 43% non-drinkers, 43% moderate drinkers, and 14% heavy drinkers, following WHO global alcohol reports and related literature.
4. Medical condition prevalence was grounded in CDC and WHO data, with approximately 51% healthy, 32% hypertension, 11% diabetes, and 6% comorbid hypertension and diabetes.
5. Condition assignment was made age-dependent, such that younger patients were predominantly healthy and chronic conditions increased with age, reflecting known epidemiological patterns.
6. Blood oxygen saturation (SpO_2) was capped at 99% with a clinically appropriate mean of 97.8% and a standard deviation of 1.0.
7. Body weight was derived from a realistic BMI distribution capped at 40 kg/m^2.

Correlated variables such as heart rate, ECG results, medication use, and health score were also adjusted to reflect plausible clinical associations with the corrected features. The use of synthetic data is justified by patient privacy constraints, and the corrected generation process is grounded in documented real-world distributions, addressing a key limitation of the original dataset.

Data Preprocessing

A preprocessing pipeline was implemented to address remaining data quality and consistency issues. Irrelevant device identifier columns (Device_id, Device_id.1, Device_id2, calorie_device, tem_dev, and muscle_dv) were removed as they contained non-informative values. Numerical features such as age, weight, and blood pressure were imputed using the median where missing, as this measure is robust to skewness and outliers. Categorical features were filled using the mode to preserve the natural distribution of categories. Remaining outliers were treated using the Interquartile Range (IQR) method, where extreme values were capped rather than removed to avoid unnecessary data loss.

Exploratory Data Analysis

The exploratory analysis of the corrected dataset revealed several meaningful and epidemiologically coherent insights, as summarised in Figure 1. The gender distribution is balanced between male and female, which minimises bias in model predictions (Figure 1a). The distribution of medical conditions reflects real-world prevalence: the majority of patients (52.4%) are healthy, while hypertension (29.6%) and diabetes (11.5%) are the most common conditions, with a smaller comorbid group (6.5%), as shown in Figure 1c. A clear age gradient was observed across conditions (Figure 1b), with healthy patients being notably younger (mean age approximately 38 years) and the comorbid hypertension and diabetes group being the oldest (mean age approximately 61 years), consistent with the known epidemiology of chronic disease onset. Analysis of lifestyle factors confirmed expected associations (Figure 1d): smoking and heavy alcohol consumption were both positively associated with hypertension and diabetes, while the healthy cohort exhibited the lowest rates of both behaviours. Heart rate, blood oxygen saturation (SpO_2), and health score all showed realistic, clinically interpretable distributions, lending credibility to downstream modelling based on this corrected dataset.